F78AA - Actuarial and Financial Mathematics A

Tutorial 5 – Yields

Question 1 You pay a price of £1 000 for an investment that will repay you £120 at the end of each of the next 10 years. Calculate the yield on this investment.

Question 2 You invest £10 000 in a project and receive back £700.87 per annum payable half-yearly in arrears for the next 25 years. Calculate the yield on this investment.

Question 3 You invest £100 000 in a project which returns £20 000 at the end of each of the first, second and third years, and then £40 000 at the end of each of the fourth, fifth and sixth years. Calculate the yield on this project.

Question 4 You invest in a mining project which requires you to pay £1 000 000 at the start of each of the calendar years 2018, 2019 and 2020 to develop the mine. It is expected to be in active production for twenty years, returning a profit of £250,000 at the end of each calendar year 2024 to 2043 inclusive. Calculate the yield on this project.

Question 5 Suppose you have two series of cashflows:

- cashflows of $a_1, a_2, \ldots, a_n$ are payable at times $t_1, t_2, \ldots, t_n$, and
- cashflows of $b_1, b_2, \ldots, b_m$ are payable at times $s_1, s_2, \ldots, s_m$,

for strictly positive integers n and m. Suppose that the yields on both sets of cashflows exist and are equal to the same value $i^\star$.

Show that if you combine the two sets of cashflows into a single transaction, then $i^\star$ is also a solution of the equation of value of the new, combined cashflows.

However, setting $n = m = 2$, find an example to show that it need not be the unique such solution ≥ -1.

Question 6 An investor is considering investing in two possible projects.

- Project A is a smartphone factory, which will cost £200 million immediately to build, and at the end of each of the first, second, third, fourth and fifth years will make a profit of £60 million. At the end of the fifth year the current generation of mobile telephones will be obsolete, and the factory will be sold for £20 million.
- Project B is identical to Project A until the end of the fifth year. At that time the factory will make a profit of £60 million as before, but instead of being sold the factory will be re-tooled at a cost of £180 million to produce the next generation of smartphones. It will then make a profit of £60 million at the end of the sixth, seventh, eighth, ninth and tenth years, and at the end of the tenth year it will be sold for £20 million.

1. Calculate the yield on Project A.
2. Calculate a yield on Project B.
3. State sufficient conditions, involving the signs of the net cash-flows under a financial transaction, under which there exists a unique yield $i^\star \geq -1$ on that transaction.
4. Verify that Project A satisfies the conditions above, but that Project B does not.
5. Prove that the yield on Project B found in part (2) above is unique, despite part (4) above.

Question 7 You have £1 000 000 to invest, and you have a choice of two investments.

- Investment A will return £130 000 at the end of each of the next 10 years, and in addition will return your original investment of £1 000 000 at the end of the 10 years.
- Investment B will return a series of payments, annually in arrears, starting at £160 000 at the end of the first year, then increasing by 5% per annum compound until the final payment is made at the end of the 10th year.

Using the highest yield as the decision-making criterion, determine the investment that you would choose.

Solutions: Q1. 0.0345; Q2. 0.0498; Q3. 0.1679; Q4 0.0344; Q6.1 0.1722; Q6.2 0.1722; Q7. B.

F78AA - Actuarial and Financial Mathematics A
Solutions to Tutorial 5 – Yields

Solution 1 The equation of value is

$$f(i) := -1\,000\,(1+i)^{10} + 120\,s_{\overline{10}|}^{i} = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. To get a starting value for the trial-and-error method, we do a first order approximation of the equation of value:

$$f(i) \approx -1\,000\,(1+10i) + 120\left(\frac{1+10i-1}{i}\right) = 0 \quad \Rightarrow \quad i \approx \frac{200}{10\,000} = 0.02.$$

We start by evaluating $f(0.02)$, $f(0.03)$, and so on until we find two values where the sign of f changes. Using this method, we calculate $f(0.03) = +31.75$ and $f(0.04) = -39.51$. We seek $i^\star$ such that $f(i^\star) = 0$ and assume that f is linear between $i = 0.03$ and $i = 0.04$, i.e.

$$\frac{i^\star - 0.03}{0.04 - 0.03} = \frac{f(i^\star) - f(0.03)}{f(0.04) - f(0.03)} \quad \Rightarrow \quad i^\star \approx 0.03 + 0.01\,\frac{31.75}{31.75 + 39.51} = 0.0345.$$

Solution 2 Using a time unit of a half-year, the equation of value is

$$f(i) := -10\,000\,(1+i)^{50} + \frac{700.87}{2}\,s_{\overline{50}|}^{i} = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. To get a starting value for the trial-and-error method, we do a first order approximation of the equation of value:

$$f(i) \approx -10\,000\,(1+50i) + \frac{700.87}{2}\left(\frac{1+50i-1}{i}\right) = 0 \quad \Rightarrow \quad i \approx \frac{7521.75}{500\,000} = 0.015.$$

We start by evaluating $f(0.015)$, $f(0.02)$, and so on until we find two values where the sign of f changes. We find $f(0.02) = +2723.702$ and $f(0.025) = -209.159$. We seek $i^\star$ such that $f(i^\star) = 0$ and assume that f is linear between $i = 0.02$ and $i = 0.025$, i.e.

$$\frac{i^\star - 0.02}{0.025 - 0.02} = \frac{f(i^\star) - f(0.02)}{f(0.025) - f(0.02)} \quad \Rightarrow \quad i^\star \approx 0.02 + 0.005\,\frac{2723.702}{2723.702 + 209.159} = 0.0246.$$

Hence in time units of 1 year, the yield is approximately 0.0498. (The exact solution is 0.05.)

Solution 3 In thousands of pounds, the equation of value is

$$f(i) := -100\,(1+i)^{6} + 20\,s_{\overline{6}|}^{i} + 20\,s_{\overline{3}|}^{i} = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. To get a starting value for the trial-and-error method, we do a first order approximation of the equation of value:

$$f(i) \approx -100\,(1+6i) + 20\left(\frac{1+6i-1}{i}\right) + 20\left(\frac{1+3i-1}{i}\right) = 0 \quad \Rightarrow \quad i \approx \frac{80}{600} = 0.013.$$

We start by evaluating $f(0.013)$, $f(0.14)$, and so on until we find two values where the sign of f changes. We find $f(0.16) = +6.022$ and $f(0.17) = -1.601$. We seek $i^\star$ such that $f(i^\star) = 0$ and assume that f is linear between $i = 0.016$ and $i = 0.017$, i.e.

$$\frac{i^\star - 0.16}{0.17 - 0.16} = \frac{f(i^\star) - f(0.16)}{f(0.17) - f(0.16)} \quad \Rightarrow \quad i^\star \approx 0.16 + 0.01\,\frac{6.022}{6.022 + 1.601} = 0.1679.$$

Solution 4 Be careful to distinguish between 'start of year' and 'end of year'! In thousands of pounds, the equation of value is

$$f(i) = -1\,000\,\ddot{s}^{i}_{\overline{3}|}\,(1+i)^{23} + 250\,s^{i}_{\overline{20}|} = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. To get a starting value for the trial-and-error method, we do a first order approximation of the equation of value in which we also approximate $d \approx i$:

$$f(i) \approx -1\,000\,\frac{1+3i-1}{i}\,(1+23i) + 250\left(\frac{1+20i-1}{i}\right) = 0 \quad \Rightarrow \quad i \approx \frac{2\,000}{69\,000} = 0.029.$$

We start by evaluating $f(0.03)$, $f(0.04)$, and so on until we find two values where the sign of f changes. We find $f(0.03) = +434.43$ and $f(0.04) = -557.09$. We seek $i^\star$ such that $f(i^\star) = 0$ and assume that f is linear between $i = 0.03$ and $i = 0.04$, i.e.

$$\frac{i^\star - 0.03}{0.04 - 0.03} = \frac{f(i^\star) - f(0.03)}{f(0.04) - f(0.03)} \quad \Rightarrow \quad i^\star \approx 0.03 + 0.01\,\frac{434.43}{434.43 + 557.09} = 0.0344.$$

Solution 5 Let $f_1(i) = 0$ and $f_2(i) = 0$ be the two equations of value associated with the first and second series of cashflows, respectively, both at time 0 rather than at time t_n and s_m, respectively. That is:

$$\begin{aligned} f_1(i) &:= a_1\,v^{t_1} + a_2\,v^{t_2} + \cdots + a_n\,v^{t_n} = \sum_{k=1}^{n} a_r\,v^{t_k} = 0, \\ f_2(i) &:= b_1\,v^{s_1} + b_2\,v^{s_2} + \cdots + b_m\,v^{s_m} = \sum_{\ell=1}^{m} b_r\,v^{s_\ell} = 0. \end{aligned}$$

and we are told that $f_1(i^\star) = 0$ and $f_2(i^\star) = 0$.

If we create a new set of cashflows simply by combining the two sets, the new equation of value at time 0 is

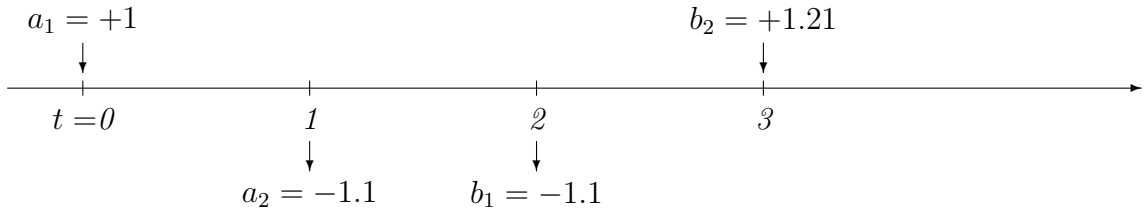
$$f(i) := f_1(i) + f_2(i) = 0.$$

Then

$$f(i^\star) = f_1(i^\star) + f_2(i^\star) = 0 + 0 = 0,$$

i.e. $i^\star$ satisfies the equation of value for the combined set.

Now we look for an example to show that $i^\star$ need not be the unique such solution ≥ -1. Consider the following sets of cashflows:



The two equations of value are:

$$f_1(i) := (1+i) - 1.1 = 0 \qquad \text{and} \qquad f_2(i) := -1.1\,(1+i) + 1.21 = 0.$$

It is easily checked that $i^\star = 0.1$ is the unique real solution ≥ -1 of each of these. Notice that $f_2(i) = -1.1\,f_1(i)$, so when we write down the equation of value of the combined series of cashflows,

$$f(i) = f_1(i) + f_2(i) = f_1(i)\left(1 - 1.1v^2\right) = 0.$$

Then for $f(i) = 0$, either $f_1(i) = 0$ or $1 - 1.1v^2 = 0$. Solving the latter equation gives a second real solution $j = 1.1^{1/2} - 1 \approx 0.0488088 \geq -1$. Hence, while there is a solution to the combined equation of value, it is not unique. Thus the yield is undefined.

Solution 6

1. The cashflows under Project A (in £million) are:

Time	0	1	2	3	4	5
Cashflow	-200	$+60$	$+60$	$+60$	$+60$	$+80$

The equation of value is:

$$f_A(i) := -200\,(1+i)^5 + 60\,s_{\overline{5}|}^{i} + 20 = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. As $f_A(0.17) = +2.374$ and $f_A(0.18) = -8.299$, and we seek $i^\star$ such that $f_A(i^\star) = 0$,

$$\frac{i^\star - 0.17}{0.18 - 0.17} = \frac{f_A(i^\star) - f_A(0.17)}{f_A(0.18) - f_A(0.17)} \quad \Rightarrow \quad i^\star \approx 0.17 + 0.01\,\frac{2.374}{2.374 + 8.299} = 0.1722.$$

2. The cashflows under Project B (in £million) are:

Time	0	1	2	3	4	5	6	7	8	9	10
Cashflow	-200	$+60$	$+60$	$+60$	$+60$	$+60-180$ $=+80-200$	$+60$	$+60$	$+60$	$+60$	$+80$

We can solve directly for the yield on Project B. However, it is quicker to note that Project B can be decomposed into two series of cashflows, in both of which the amounts of cashflow are the same as for Project A.

Hence the equation of value for Project B can be written as:

$$f_B(i) := f_A(i)\,(1+i)^5 + f_A(i) = 0 \qquad \Rightarrow \qquad f_B(i) := f_A(i)\left((1+i)^5 + 1\right) = 0.$$

To solve $f_B(i) = 0$, we solve either $f_A(i) = 0$ or $(1+i)^5 + 1 = 0$. From the last part of the question, $f_A(0.1722) \approx 0$. So a yield on Project B is 0.1722.

3. From a result in the lecture notes, a series of cashflows has a well-defined yield (a unique real solution ≥ -1 of the equation of value) if all the positive cashflows precede all the negative cashflows, or *vice versa*.

4. It is clear that Project A satisfies the conditions in part (3) above, but that the net cashflow of -120 at $t = 5$ under Project B means that Project B does not.

5. From part (2) above, we can write:

$$f_B(i) = f_A(i)\,\left((1+i)^5 + 1\right) = 0.$$

As $i^\star$ is a well-defined yield for Project A, any real solution ≥ -1 other than $i^\star$ must be a solution of $(1+i)^5 + 1 = 0$. But this equation has no real roots ≥ -1. Hence $i^\star$ is indeed the yield on Project B.

Solution 7 The yield on Investment A is 13% p.a. effective.

In thousands of pounds, the equation of value for Investment B (in £000s) is

$$\begin{aligned} f(i) &:= -1\,000\,(1+i)^{10} + 160 \sum_{k=1}^{10} 1.05^{k-1}\,(1+i)^{10-k} = 0 \\ \Rightarrow 0 &= -1\,000\,(1+i)^{10} + 160 \times 1.05^{9} \times \sum_{k=1}^{10} 1.05^{k-10}\,(1+i)^{10-k} \\ &= -1\,000\,(1+i)^{10} + 160 \times 1.05^{9} \times \sum_{k=1}^{10} \left(\frac{1+i}{1.05}\right)^{10-k} \\ &= -1\,000\,(1+i)^{10} + 160 \times 1.05^{9} \times s_{\overline{10}|}^{i'}, \end{aligned}$$

where $i' := (1+i)/(1.05) - 1$.

As we cannot solve the equation of value exactly, we use trial-and-error. As $f(0.13) = +136.778$ and $f(0.14) = -12.418$, and we seek $i^\star$ such that $f(i^\star) = 0$,

$$\frac{i^\star - 0.13}{0.14 - 0.13} = \frac{f(i^\star) - f(0.13)}{f(0.14) - f(0.13)} \quad \Rightarrow \quad i^\star \approx 0.13 + 0.01\,\frac{136.778}{12.418 + 136.778} = 0.1392.$$

Hence you would choose Investment B as it results in the highest yield.